Predicting SME Loan Default Risk in Kenya

Using Gradient Boosting & Explainable AI

A machine learning approach to early loan default detection

Applied Research Project (DSC 200) | School of Science & Technology

OPEN UNIVERSITY OF KENYA

PRESENTATION DETAILS

Course: Data Science
Unit: Applied Research Project [DSC 200]
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Project Goals & Stakeholders

- **Objective: predict early loan defaults using machine learning.**
- Reduce non-performing loans efficiently.
- Intervene early with high-risk borrowers.
- Promote financial inclusion through fairer credit decisions.
- Primary stakeholders: credit officers and risk analysts.
- **Secondary stakeholders: SME lenders and regulators.**

Data Characteristics Overview

- **Source: 277,159 loan records from Micromart Africa's digital lending platform.**
- Collected for the purpose of loan default prediction.
- Contains borrower demographics and loan terms.
- Structured as a binary classification problem.
- **Target variable: whether the borrower defaulted on the loan.**

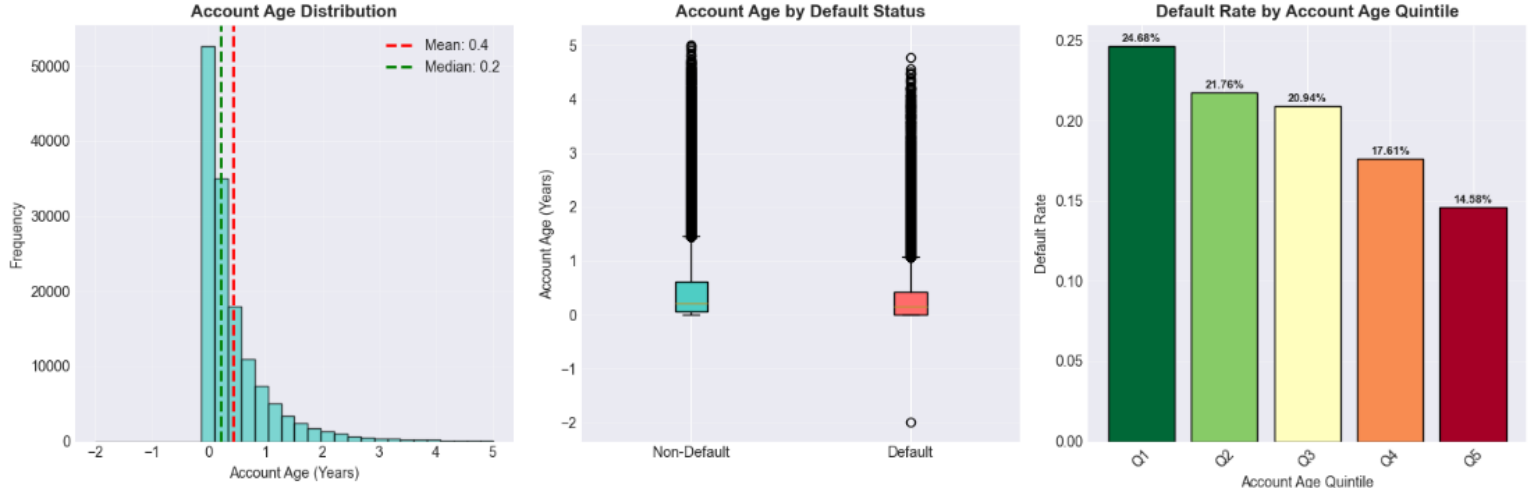
Data Cleaning & Preparation

- **Cleaning covered missing values, standardising variables and ensuring accurate entries.**
- To reduce noise and avoid data leakage, three groups of columns were removed:
- 1 · Identifiers and system references, e.g. loanid
- 2 · Highly missing features, e.g. graduationcount
- 3 · Operational and administrative fields, e.g. approvalstage
- **The model now focuses on meaningful borrower, loan and repayment-behaviour features that generalise well for credit-risk prediction.**

Insights from Exploratory Data Analysis

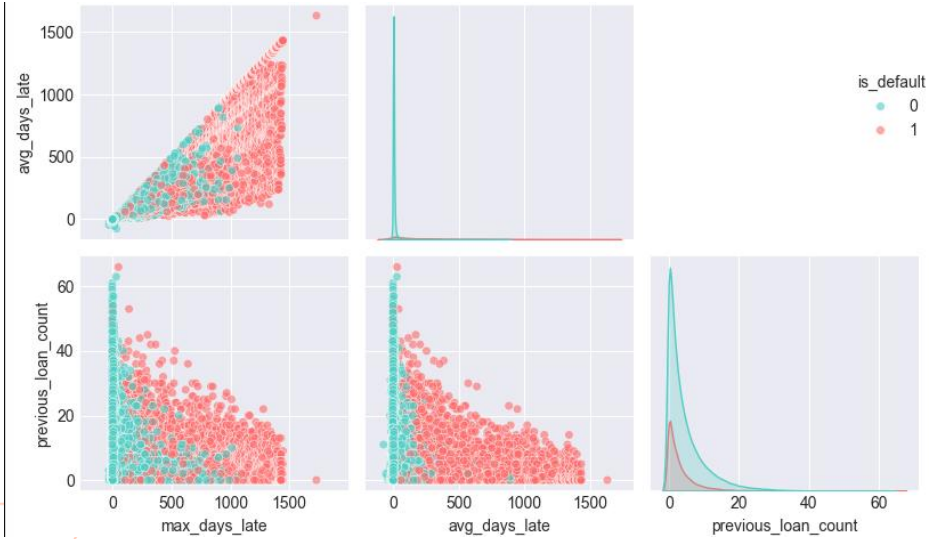
- **Younger entrepreneurs face significantly higher default rates.**
- **Missed payments strongly correlate with eventual default.**

Age vs. Default



- Default risk decreases with age.
- Younger borrowers are significantly riskier.
- Correlation with default is -0.07 (weak negative).

Top 3 Variables vs. Default



- High collinearity between max_days_late and avg_days_late is visually evident.
- Defaulters have significantly higher days late — clear separation in the scatter plots.
- previous_loan_count shows weaker discriminatory power visually.
- Non-linear boundaries may exist — the default / non-default separation isn't perfectly linear.

Data Preparation Overview

01

Scaling

Features were scaled for uniformity, helping algorithms converge faster and more accurately.

02

Encoding

Categorical variables were converted to numeric formats so algorithms can interpret them.

03

Train-Test Split

The data was divided into training and testing sets to evaluate the model and confirm it generalises to unseen data.

Why it matters: **Scaling speeds up model convergence, encoding lets algorithms read categories, and the split lets us test on unseen data — so the performance results can be trusted.**

Baseline Model Performance

01

Recall: 92.53%

Captures 92.5% of actual defaults — good for early warning.

02

Precision: 97.64%

When the model predicts default, it is correct 97.6% of the time.

03

F1-Score: 95.01%

A strong balance between precision and recall.

04

Accuracy: 98.07%

Overall prediction accuracy on the test set.

Why recall leads: **Recall is the most critical metric because we want to catch as many potential defaults as possible (fewest false negatives). Precision is secondary: some false positives are acceptable if more true defaults are caught.**

Advanced Models

- **Two advanced models were chosen to go beyond the baseline:**
- Random Forest — captures non-linear interactions, is robust to noisy features and is strong with behavioural data.
- LightGBM — employed to model complex non-linear interactions in repayment behaviour.

Model Comparison: LightGBM vs. Previous Models

Logistic Regression

Baseline: recall 92.53%, F1 95.01%. Slowest to train at 13 minutes.

Random Forest

Ensemble of trees — trained in about 2 minutes.

BEST MODEL

LightGBM

95.84%

RECALL

97%

F1-SCORE

Training time: Logistic Regression 13 min · Random Forest 2 min · LightGBM 1 min. LightGBM is fastest, balanced (F1 97%, accuracy 98%) and shows minimal overfitting.

Preferred Model: LightGBM

- **Why LightGBM is superior:**
- 1 · Highest default detection rate (95.84% recall)
- 2 · Best overall performance balance
- 3 · Computational efficiency
- 4 · Modern algorithm advantages
- **Cross-validation confirms LightGBM is robust and reliable for early loan default detection: high recall (98.70%) catches potential defaults, while excellent precision (99.64%) ensures operational efficiency.**

Recommendations & Conclusion

1	**ML-Based Credit Scoring**	Adopt machine-learning credit scoring instead of relying only on manual credit assessment.
2	**Risk-Based Segmentation**	Segment borrowers by risk: offer lower interest rates and limit exposure to high-risk borrowers.
3	**Early Warning System**	Use the model as an early warning system to reduce non-performing loans.
4	**Human Decision-Making**	Integrate the model with human judgement — it supports credit officers rather than replacing them.
5	**Continuous Improvement**	Retrain with new data over time so the model stays relevant.

Thank You for Your Attention

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